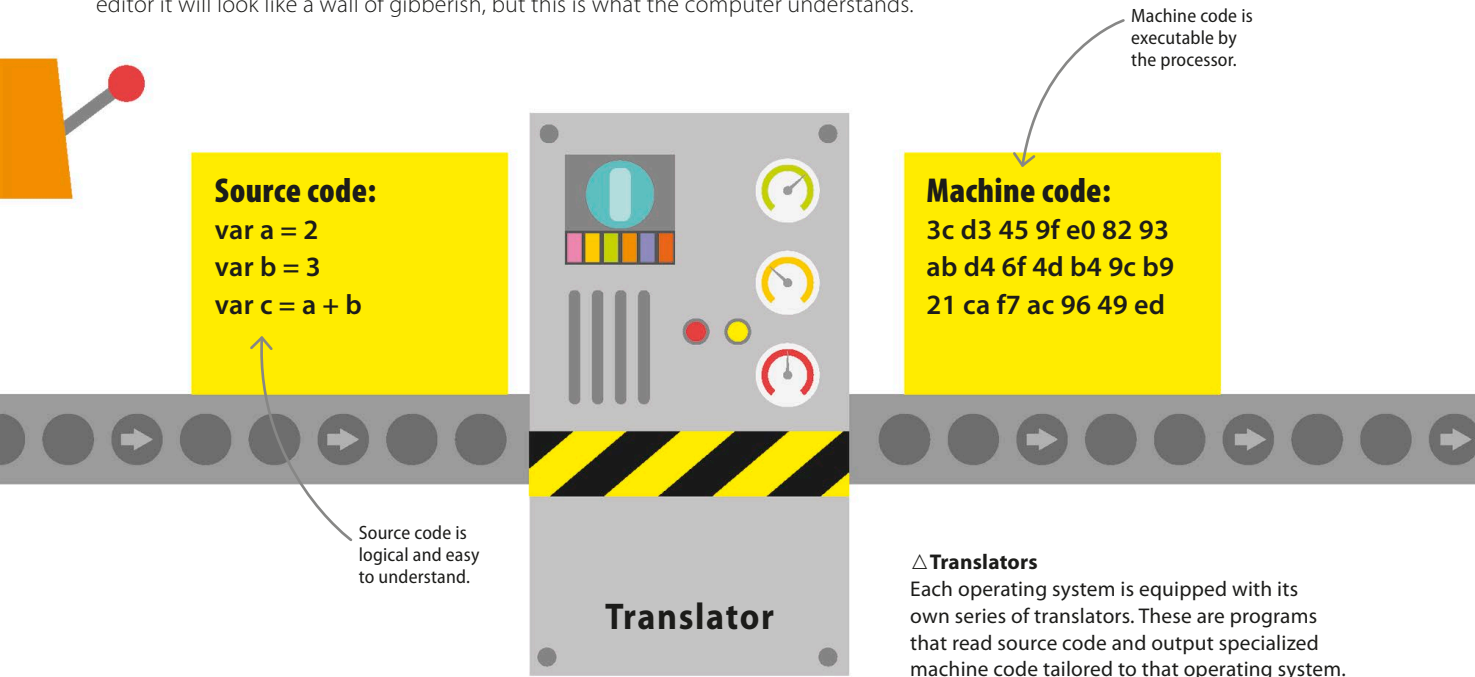


Source code vs machine code

Programmers write source code in languages like Java, Python, or C. These files contain the human-readable instructions that create individual programs, such as text editors, web browsers, or multimedia games. However, in order for the computer to run the code, the instructions must be translated into raw bits that a CPU can process. This is called machine code. If you open machine code in a text editor it will look like a wall of gibberish, but this is what the computer understands.



Opcodes and operands

In machine code, each instruction is composed of an opcode (short for operational code) and one or more operands. An opcode is a number that corresponds to a specific CPU action. For instance, the opcode 04 could mean add two pieces of data. To make opcodes more intuitive they're given standardized nicknames, also known as mnemonics. Operands are the data being processed.

▽ Machine-level programming

Programming at the machine level requires programmers to memorize the opcodes of a particular operating system. Sometimes, they even work with individual bits.

Opcode	Mnemonic	Binary	Description
87	ADD A	10000111	Add contents of register A to that of the accumulator
3A	LDA	00111010	Load data stored in the given memory address
79	MOV A C	01111001	Move data from register A to C
C3	JMP	11000011	Jump to instructions in specified memory address
C1	POP B	11000001	Pop from stack and copy to memory registers B + C